Oxford Alpha Fund

Statistical Factor Relative Value in Natural Gas Futures

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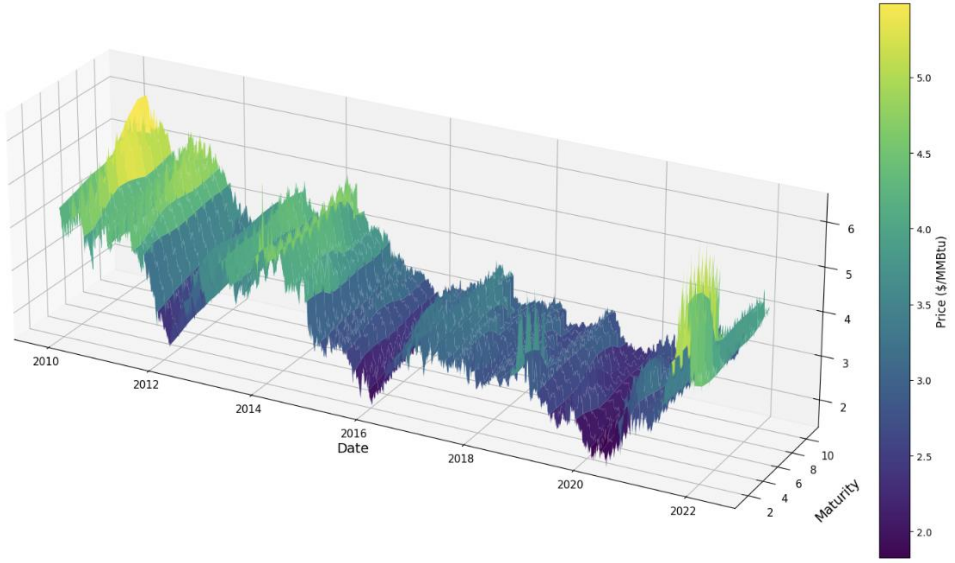

Investment Thesis

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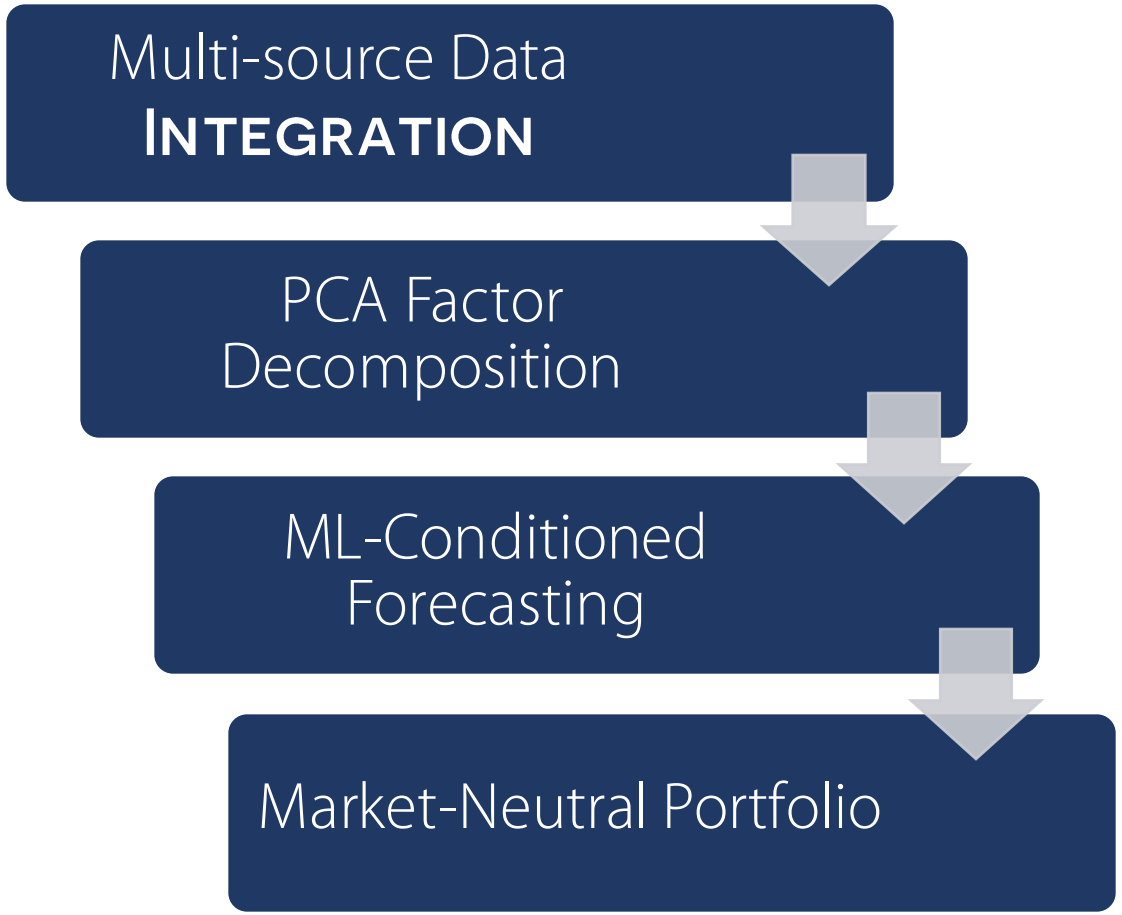
Statistical relative value on the Henry Hub futures curve

Strategy Objective: Build a market-neutral relative-value strategy on the NYMEX Henry Hub forward curve, trading constant-maturity contracts M1–M12. The objective is to identify which maturities are temporarily rich or cheap, rather than forecasting the outright gas price.

Alpha Hypothesis: Maturity-specific dislocations from the PCA-implied forward curve are partially predictable. These dislocations may become more predictable when storage constraints, weather shocks, and curve-state signals indicate stress in the natural gas market.

Core Mechanism: A rolling PCA removes common curve factors to isolate idiosyncratic residuals. A machine learning model then forecasts 5-day cumulative residual returns using storage, weather, curve-shape, and macro features. The resulting portfolio is dollar-neutral and factor-neutral.



Data Universe

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Prices — Henry Hub Futures

NYMEX Henry Hub constant-maturity futures, daily settlements, 2010–2026 (LSEG).

Estimation Universe: M1-M12

Log-returns are roll-adjusted to remove mechanical month-end continuation bias.

Storage — EIA Weekly

U.S. underground working gas in storage, reported weekly by the EIA.

Captures the **supply buffer** of the gas market and the inventory-driven **convexity** of the futures curve.

Weather — HDD / CDD

NOAA/CPC daily population-weighted CONUS temperature.

HDD and CDD proxy weather-driven gas demand:

HDD = max(65°F – T, 0) **CDD** = max(T – 65°F, 0)

Heating demand dominates winter risk; cooling demand captures summer power burn pressure.

Macro Controls

Daily FRED and CBOE series capturing cross-asset, funding, and risk conditions.

WTI crude, TTF, EUR/USD, 3M T-Bill, S&P 500, VIX, OVX.

Sources: LSEG (NYMEX HH futures), EIA (weekly storage), NOAA/CPC (HDD/CDD CONUS), FRED & CBOE (macro).

Factor Framework: Overview

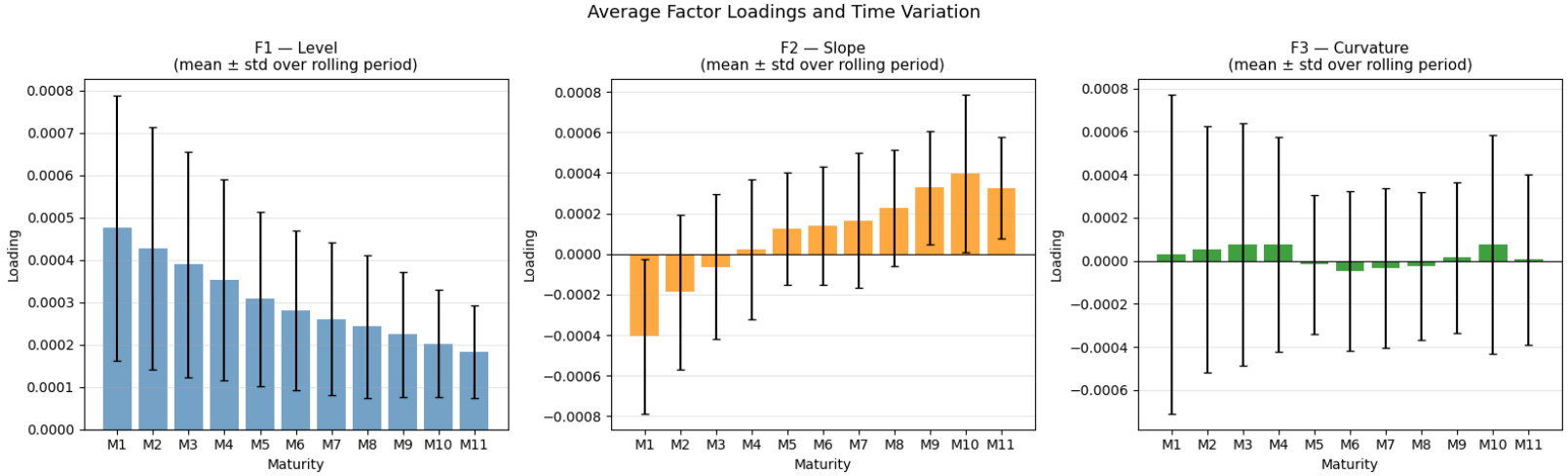
Rolling PCA isolates common curve moves and leaves maturity-specific dislocations.

Objective:

- Separate systematic curve moves from relative-value dislocations.
- A residual ϵ_i for each maturity, used as the tradable signal.

Three interpretable latent factors:

- Level — parallel movement of the curve
- Slope — front vs. back tilt
- Curvature — belly distortion



The factor model transforms raw curve moves into a cleaner cross-sectional residual signal.

Factor Framework: Two-Stage PCA

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A rolling two-stage PCA extracts stable curve factors while controlling for maturity-specific volatility.

Why two stages?

•1. Recent data matters more: Rolling exponential weights adapt to regime shifts.

•2. Maturities have different volatility: Front contracts are much noisier than back contracts.

•3. Reweighting prevents noisy maturities from dominating PCA. The final factors capture curve structure, not volatility hot spots.

Stage 1 — Estimate idiosyncratic volatility

- A fast decay weighted PCA is used to remove the first common curve components. The residual variance gives one volatility proxy per maturity.

Stage 2 — Reweighted PCA/SVD

- Returns are scaled by inverse idiosyncratic volatility and slow time weights.

$$\widetilde{R}_t = W_{\sigma,t}^{-1} R_t W_{\text{time}}^{1/2}$$

Statistical Factor Model :

- daily returns=common curve factors + maturity-specific residual

$$r_t = B_{t|t-1} f_t + \varepsilon_t$$

- rt: daily log-returns of M1–M12
- Bt: factors loadings
- ft: factor returns, WLS-fit at each date t
- εt: idiosyncratic residual — the maturity-specific dislocation

- Estimate rolling loadings B from past returns:

$$\widetilde{R}_t = U_t S_t V_t^\top \qquad B_{t|t-1} = W_{\sigma,t} U_{t,m}$$

- Then project each new return vector onto the factor space.

$$f_t = \text{WLS}(r_t \sim B_{t|t-1})$$

[1]: *Paleologo, The Elements of Quantitative Investing — Statistical Factor Model*

Target Variable

Predicting standardized residual returns aligns the ML target with a factor-neutral relative-value strategy.

Why residuals?

The objective is not to forecast outright gas direction, but to rank maturities by expected residual outperformance.

Raw returns mix outright gas direction, common curve shocks, and maturity-specific dislocations.

Residual returns remove Level, Slope, and Curvature, leaving the component aligned with a factor-neutral long/short strategy.

Target Standardization:

$$z_{i,t} = \frac{y_{i,t}^{raw}}{\sqrt{5}\,\hat{\sigma}_{\varepsilon,i,t}^{63d}}$$

- The target is standardized by trailing residual volatility, using information available at date t only.
- The target is expressed in residual-risk units to make maturities comparable.

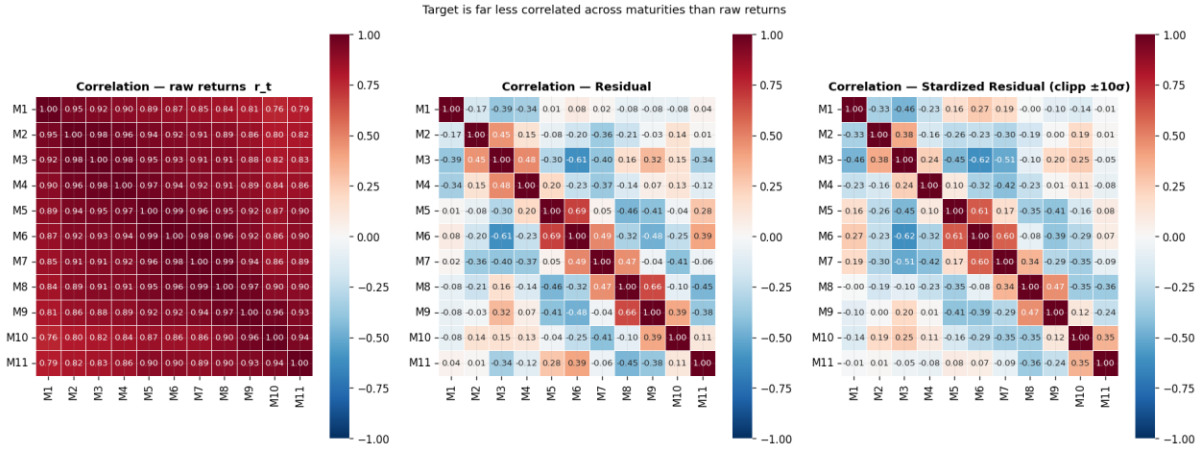
Trading interpretation:

- $z > 0$ ⟶ maturity expected to outperform the factor-implied curve
- $z < 0$ ⟶ maturity expected to underperform the factor-implied curve

Prediction target: 1-week cumulative residual return

Because log-returns are additive, summing daily residuals gives the next-week residual move.

$$y_{i,t}^{raw} = \sum_{h=1}^{5} \varepsilon_{i,t+h}$$



Feature Engineering

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~70 features grouped into 7 signal families — per-maturity (i,t) and date-level (t).

Family	Scope	Description
Curve Shape	Per maturity	Local slope and curvature at each point on the forward curve; deviation of contract from the curve mean; change in local slope.
PCA Residual Signal	Per maturity	Current idiosyncratic dislocation level and z-score; short-term mean-reversion tendency; residual relative to the cross-sectional average.
Factor Dynamics	Date-level	Level, slope, and curvature momentum and realized vol; explained variance ratio as a regime indicator.
Fundamental — Storage	Date-level	EIA weekly inventory vs. 5-year seasonal norm; unexpected weekly change; EIA announcement days.
Fundamental — Weather	Date-level	HDD / CDD deviation from 4-year climate baseline; ARMAX expected vs unexpected temperature; 5-day forecast shock vs seasonal average.
Macro & Cross-Asset	Date-level	Crude oil momentum; energy volatility regime; equity risk sentiment (VIX); funding conditions (T-Bill); TTF–Henry Hub & JKM-HH spreads.
Calendar	Per-Maturity	Month-of-year seasonality; injection vs withdrawal season flag; contract position along the curve

[2] Y. Chen, Temperature, storage, and natural gas futures prices.

[3] M. Boons & M. Porras Prado, Basis-Momentum, J. Finance.

Training Strategy & Model Results

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Walk-forward model selection with purged validation and out-of-fold IC evaluation.

Protocol

- **Preprocessing:** Preprocessing: cross-sectional robust z-score by date; trailing z-score for time-series features.
- **Walk-forward CV** with rolling 3-year training windows and quarterly refits.
- **Purge gap** of h = 6 days to remove overlap induced by the 5-day cumulative target (break serial autocorrelation).
- In-sample 2013–2022: model selection, feature selection, hyper-parameter tuning and portfolio sizing.
- Out-of-sample 2022–2026: held out for final evaluation; no parameter touched after the cut-off.
- Feature selection based on MDI and permutation importance.

Model	IC	ICIR	Tail IC	IC t-stat	IC Gap
Ridge	0.0386	0.1060	0.0469	4.12	0.0609
Lasso	0.0391	0.1076	0.0493	4.19	0.0601
Random Forest	0.0607	0.1586	0.0663	6.17	0.1183
LightGBM	0.0699	0.1757	0.0758	6.83	0.2545
XGBoost	**0.0729**	**0.1882**	**0.0771**	**7.32**	**0.0537**

Table 1: Out-of-Fold Model Performance

IC measures the rank correlation between predicted and realized residual returns. ICIR is the information coefficient divided by its time-series volatility. Tail IC is computed on the most extreme predicted long/short opportunities. IC Gap measures the train-validation IC difference and is used as an overfitting diagnostic. The selected model is a heavily regularized XGBoost specification, consistent with the hypothesis that non-linear interactions between storage, weather and curve-state variables help predict residual mean reversion.

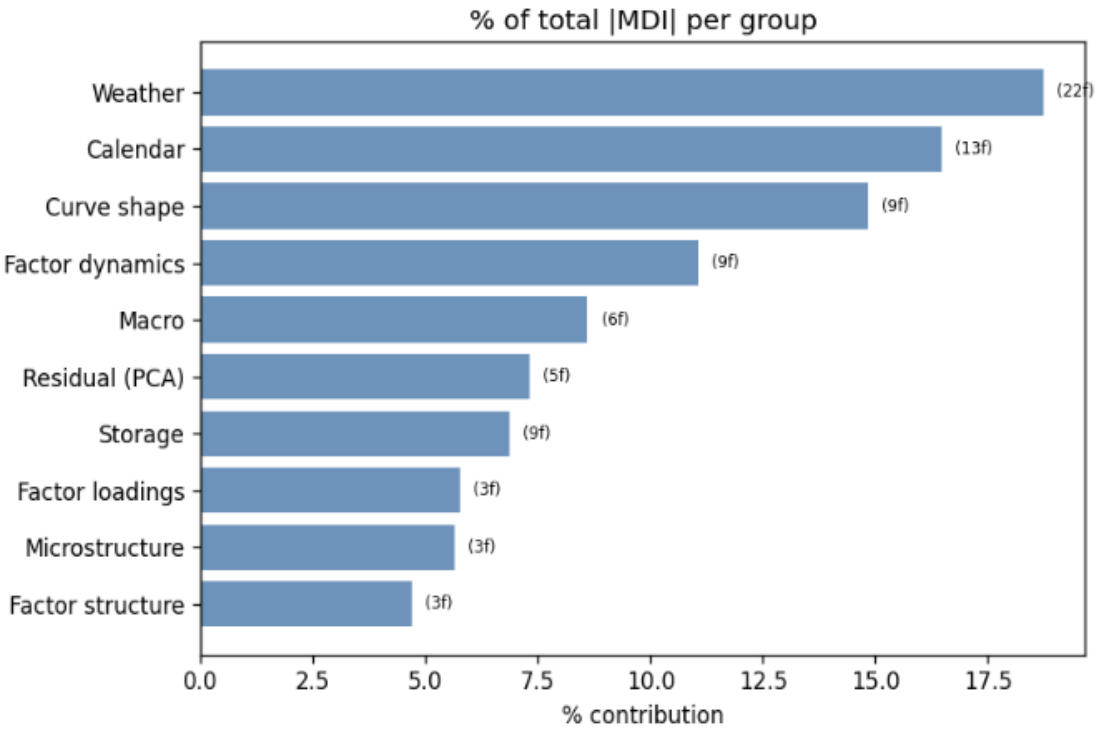
Take-away:

- The performance over Ridge/Lasso supports the **non-linearity hypothesis.**

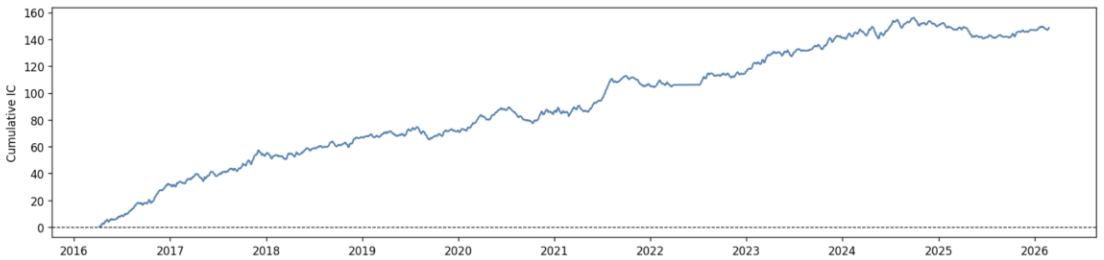
Model Analysis

Predictive power is concentrated in economically interpretable regimes and feature groups.

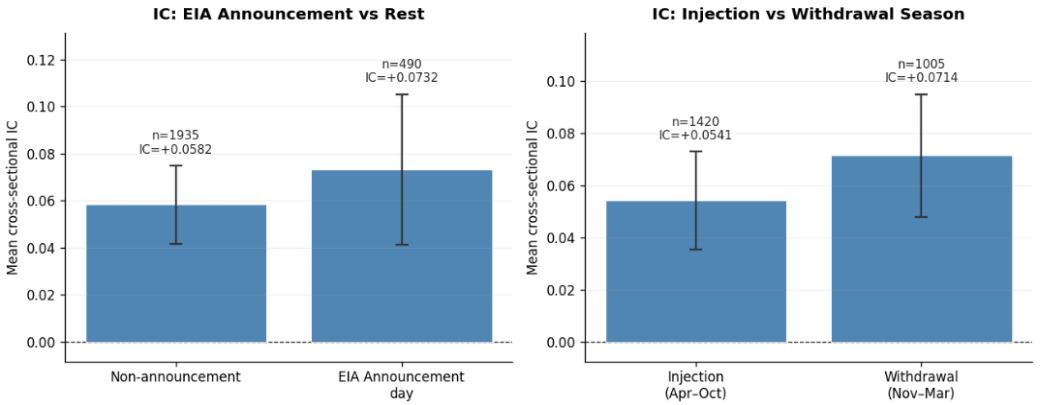
- Weather, calendar and curve-shape variables dominate model importance.



- Cumulative IC is upward-sloping, indicating persistent ranking skill.



- Signal is stronger during withdrawal season and around EIA announcements.



Portfolio Construction & Backtesting

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Rank-based relative-value portfolio, factor-neutralization and explicit transaction costs.

Portfolio Construction

1. Universe

- M2–M12 constant-maturity contracts; M1 excluded to reduce roll noise.

2. Ranking signal

- Rank maturities by predicted standardized residual return.

3. Initial long/short book

- Long top 4 maturities, short bottom 4 maturities, equal-weighted.

4. Risk controls

- Gross exposure = 1.0; max position size = 25%.

5. Neutralization

- Enforce dollar-neutrality and remove exposure to the first 3 PCA factors.

$$\widetilde{B}_t = \begin{bmatrix} B_t & \frac{\mathbf{1}}{\sqrt{N}} \end{bmatrix}, \qquad w_t^{neut} = w_t^{raw} - \mathbf{Proj}_{\widetilde{B}_t}\left(w_t^{raw}\right)$$

$$B_t^\top w_t^{neut} \approx 0, \qquad \mathbf{1}^\top w_t^{neut} \approx 0$$

Protocol

Execution Timing (no lookahead)

- Signal computed at close t --> Portfolio executed at close t+1.
- Weights are lagged before PnL is computed.

Overlapping Sleeves (H = 5 days)

- Each daily signal is held for 5 days; the live portfolio averages the 5 most recent sleeves.

$$w_t^{live} = \frac{1}{5}\sum_{h=0}^{4} w_{t-h}^{signal}$$

Transaction Costs

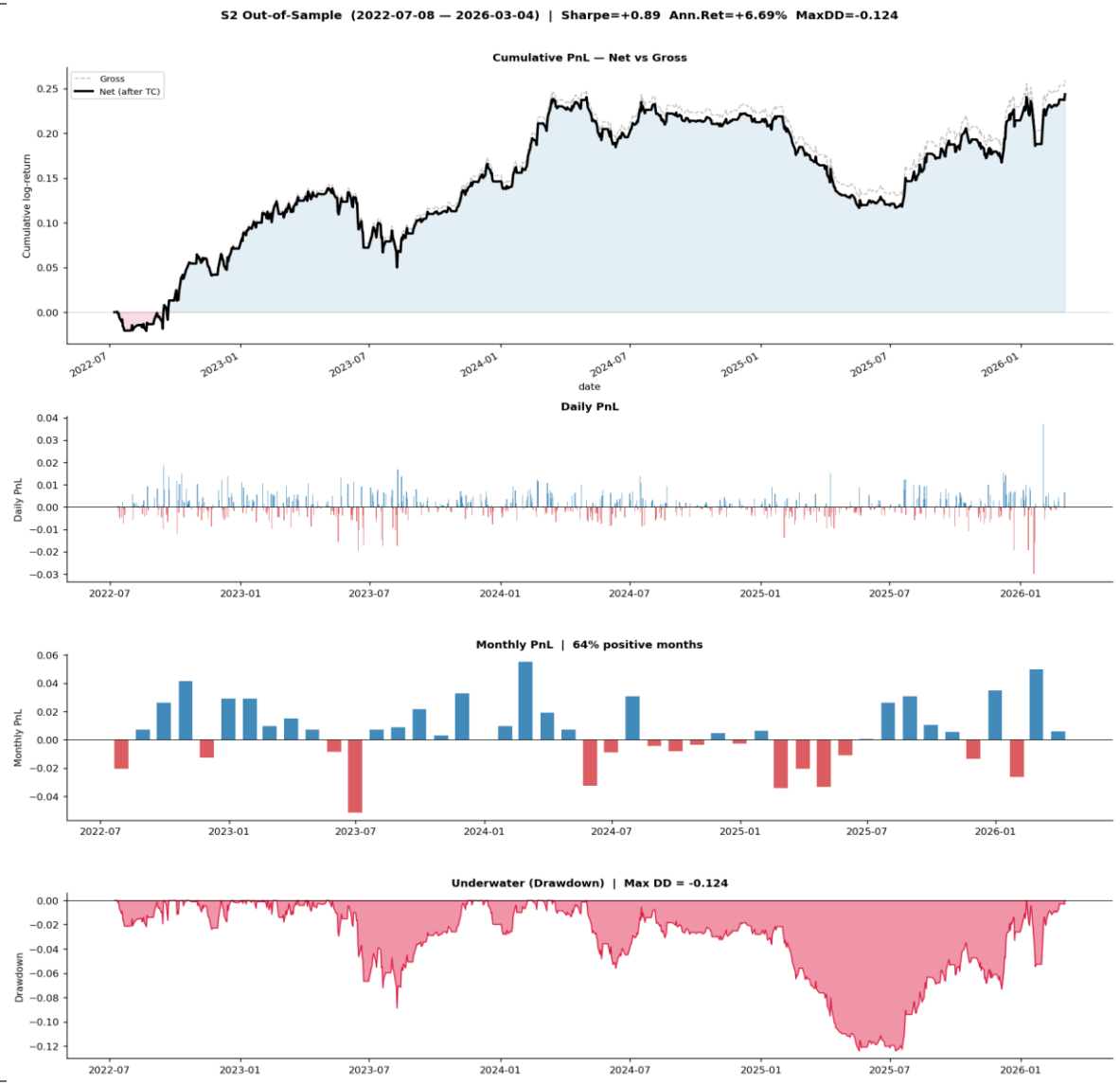
- 2 bps one-way per unit of notional traded
- Costs are charged on changes in live portfolio weights.

$$\mathrm{TC}_{t+1} = c\sum_i \left|w_{i,t}^{live} - w_{i,t-1}^{live}\right|$$

Out-of-sample Backtesting Results

July 2022 – March 2026 · 2 bps one-way transaction cost

Metric	Full OOS
Ann. Return (net)	+6.69%
Ann. Return (gross)	+7.13%
Ann. Volatility	7.56%
Ann. TC Drag	0.44%
Sharpe Ratio (net)	+0.885
Sharpe Ratio (gross)	+0.943
Sortino Ratio	+1.166
Max Drawdown	-12.4%



Limitations & Next steps

The backtest provides encouraging OOS evidence of a residual-ranking signal after costs and factor neutralization, but further robustness work is needed.

Key Limitations

1. Structural break sensitivity

The model struggles during extreme market regimes. During COVID-19, highly correlated curve moves reduced cross-sectional dispersion and weakened residual signals.

2. Roll-date contamination

Roll adjustment significantly reduces continuation-series bias but end-of-month returns remain high.

3. Simplified portfolio construction

Current portfolio is rank-based. It does not yet adapt to contract-level volatility, liquidity or transaction-cost differences.

4. Overfitting risk

Although the selected XGBoost model has a low IC gap, the signal weakens in some sub-periods, suggesting regime-dependence and the need for further robustness testing.

Next Steps

1. Validate signal robustness

Test alternative horizons $H \in \{3, 5, 10\}$, roll-exclusion windows.

2. Rebuild on individual contract data

Move from constant-maturity continuation series to true tradable futures contracts to remove residual roll uncertainty.

3. Improve portfolio optimization

Replace equal-weight top/bottom selection with volatility-scaled or constrained optimization, including factor neutrality, liquidity caps and turnover penalties.

4. Missing physical flow information

Recent supply/demand dynamics are only partially captured. LNG flows, pipeline flows, power generation and production data could improve regime identification.

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Questions?

Thank you for listening

References

[1] G. Paleologo (2025). Statistical Factor Model (Two-stage PCA). The Elements of Quantitative Investing

[2] Y. Chen (2023), Temperature, storage and Natural Gas futures prices, Journal of Futures Market, Wiley

[3] M. Boons & M. Porras Prado (2019), Basis-Momentum, Journal of Finance, Haas School of Business

[4] M. Lopez de Prado (2018), Advances in Financial Machine Learning, Wiley Finance Series

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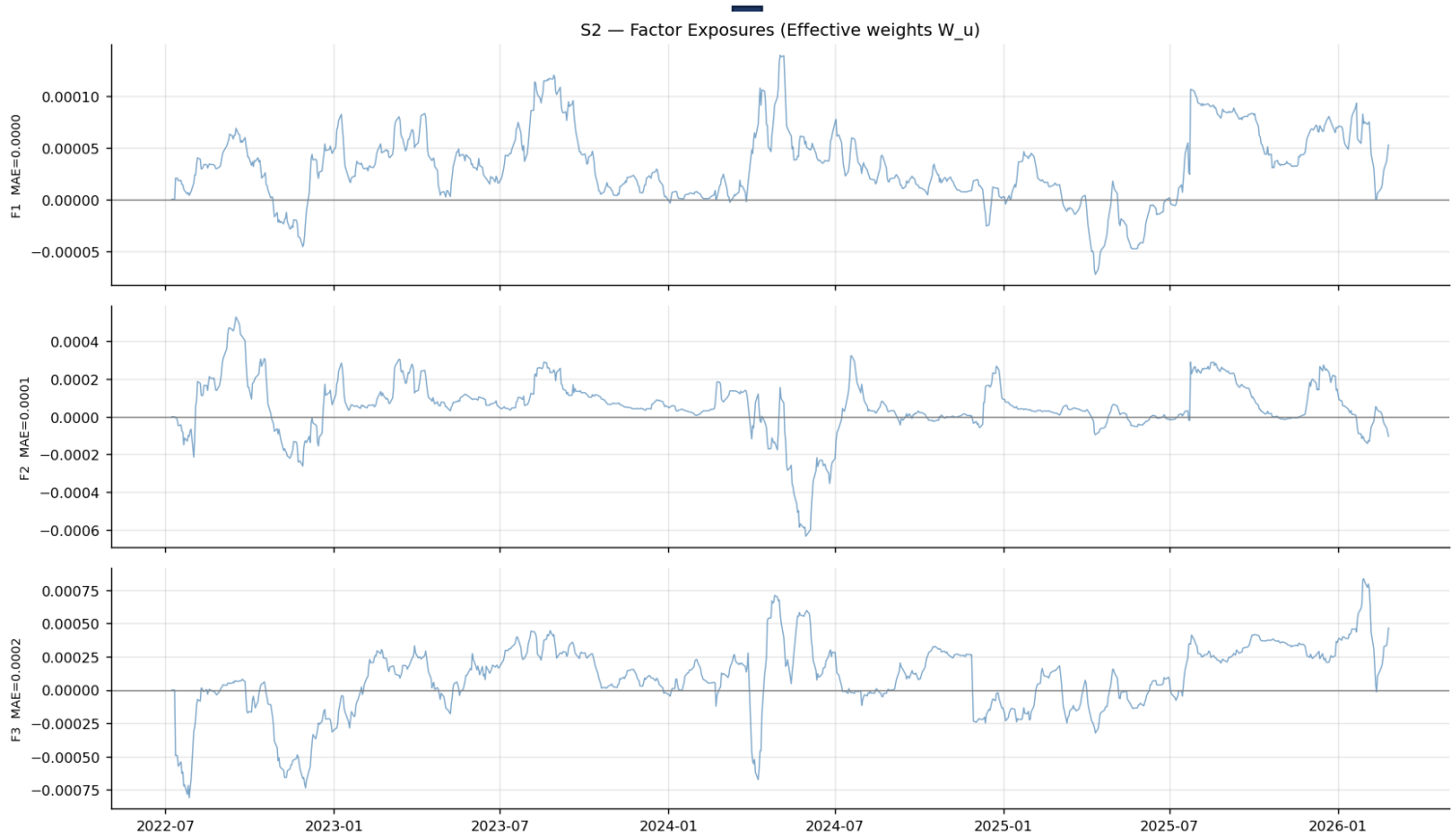
S2 — Factor Exposures (Effective weights W_u)
F1 MAE=0.0000
0.00010
0.00005
0.00000
−0.00005
F2 MAE=0.0001
0.0004
0.0002
0.0000
−0.0002
−0.0004
−0.0006
F3 MAE=0.0002
0.00075
0.00050
0.00025
0.00000
−0.00025
−0.00050
−0.00075
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2023-01
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